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ARTICLE



Intraspecific variation in seedling drought tolerance and associated traits in a critically endangered, endemic Hawaiian shrub

Andrea C. Westerband^{a,b}, Lalasia Bialic-Murphy^{a,c}, Lauren A. Weisenberger^d and Kasey E. Barton^a

^aDepartment of Botany, University of Hawaii at Manoa, Honolulu, HI, USA; ^bDepartment of Biological Sciences, Macquarie University, Macquarie Park, Australia; ^cEcology and Evolutionary Biology, University of Tennessee, Knoxville, TN, USA; ^dU.S. Fish and Wildlife Service, Pacific Islands Fish and Wildlife Office, Honolulu, HI, USA

ABSTRACT

Background: Climates are changing at a rate that exceeds the adaptive capacity of species, especially endangered species. Genetic variation and phenotypic plasticity are important for population persistence, yet few studies have linked traits to seedling performance under drought in endangered species.

Aims: We assessed intraspecific variation and trait plasticity under drought, to understand how an endangered species would cope with increasingly severe droughts.

Methods: Using greenhouse experiments, we assessed drought tolerance of a critically endangered Hawaiian shrub, *Schiedea obovata*. Seedlings from five maternal families (three selfed and two interpopulation crosses) were subjected to daily watering or a simulated drought, and we measured growth, traits linked to drought tolerance, and days until death under terminal drought.

Results: Drought reduced growth but not days until death. We detected genetic variation in growth, but no trait plasticity except in carbon:nitrogen, which decreased under drought. We did not detect traits that enhanced performance under drought but identified four physiological traits whose effects on growth varied under control and drought.

Conclusions: Our results indicate moderate drought tolerance of an endangered shrub, and low trait plasticity. Conservation of endangered species under shifting climates will benefit from studies of stress tolerance, particularly at the vulnerable seedling stage.

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Introduction

Threatened by habitat loss, invasive species, and climate change, many native plant species face an uncertain future. To persist under changing climate, plants must either tolerate new climates via phenotypic plasticity and/or *in situ* adaptation, or they must disperse to suitable habitats (Christmas et al. 2016). As climates are changing at rates that exceed the generation times and dispersal distances of most woody plant species (Davis and Shaw 2001; Christmas et al. 2016; Birks 2019), local adaptation and phenotypic plasticity are likely to be especially important for future plant persistence. Endangered plants are particularly vulnerable because their small populations may lack the genetic diversity required for adaptation to shifts in climate (Leimu et al. 2006) and because they often have limited ability to disperse into new habitats (Pegtel 1998). Unfortunately, by the very nature of their small population sizes, experimentation and research on rare plants are often limited, leading to a lack of robust climate tolerance data. Moreover, species distribution models used to predict climate change

effects on plant populations often assume implicitly that species are homogeneous throughout their distributions (Elith and Leathwick 2009). On the contrary, most species examined have demonstrated intraspecific variation across climatic gradients consistent with local adaptation (Epperson 2003; Alberto et al. 2013), and even rare plant species with limited population sizes can vary across climatic gradients (McKay et al. 2001).

Conservation of rare, threatened, and endangered plant species requires an approach that explicitly investigates intraspecific variation in plant performance with respect to microsite-scale climatic variability (Phillips-Mao et al. 2016). To obtain these data, it is essential to identify climatic variables linked to plant performance that are predicted to shift under future climate change scenarios, to perform planned crosses through breeding programmes, and then to experimentally test performance of the resulting progeny across a meaningful climatic gradient (Edwards 2015). Using these data, managers can more effectively develop translocation and mitigation actions that identify plants best suited for future habitats,

thus providing the best chance to restore rare plant species subjected to a changing climate.

Globally, conservation priorities include biodiversity hotspots such as islands with high levels of endemism. Island floras are highly threatened by invasive species, land use change, and increasingly, climate change, and now harbour some of the highest numbers of threatened and endangered species (Caujapé-Castells et al. 2010). For tropical islands, shifts in precipitation and sea level rise pose the most serious climate change threats (Harter et al. 2015). For example, the Hawaiian Archipelago has already experienced an increasing number of consecutive days without rain over the past three decades (Chu et al. 2010). Future climate change models downscaled for Hawaii project continued increases in the number of days with no rain, less precipitation overall (Timm et al. 2015; Frazier 2016; Frazier and Giambelluca 2017), and more extreme precipitation events (Chu et al. 2010; Chen and Chu 2014; Zhang et al. 2016). In Hawaii, 31% of native plant taxa are currently listed as threatened or endangered, over half are considered at risk or of conservation importance, and 10% are already extinct (Sakai et al. 2002; Mitchell et al. 2005; Weisenberger and Keir 2014). A climate-change vulnerability assessment of the native Hawaiian flora (Fortini et al. 2013) has identified species lacking future climate envelopes that were compatible with their current climate envelopes as being of particular concern, and species associated with dry forests were the most vulnerable to climate change, while generalists were among the least vulnerable. The Hawaiian Islands thus offer a compelling location to examine intraspecific variation in tolerance linked to climate change, particularly drought stress, in rare plant species.

Seasonal variability in rainfall can impose significant drought stress on plants, even in tropical mesic and wet forests (Corlett 2016; Fortunel et al. 2016). Because of their small size, limited carbohydrate reserves, and shallow rooting depth, seedlings may be particularly vulnerable to drought, and mechanisms of drought tolerance are expected to differ for seedlings compared to adult plants (Canham et al. 1999). Drought stress has been shown to reduce seedling recruitment (Engelbrecht et al. 2005; Allen et al. 2010; O'Brien et al. 2014), thereby altering tropical plant species distributions (Kursar et al. 2009). For endangered species, drought stress may further reduce already low population sizes and constrain population growth through reductions in seedling establishment (Bernardo et al. 2016; Zhang et al. 2017).

Within the native Hawaiian flora, relatively little is known about drought tolerance in seedlings, although drought has been implicated in the population declines of the threatened Haleakala silversword, *Argyroxiphium sandwicense* (Krushelnicky et al. 2013), the endangered focal species of this study, the endemic shrub, *Schiedea obovata* (Caryophyllaceae) (Bialic-Murphy and Gaoue 2018), as well as a congener *S. adamantis* (Sakai et al. 2017). Among the native Hawaiian genera, *Schiedea* is considered the most vulnerable to climate change (Fortini et al. 2013).

In this study, we used the norm of reaction approach (Simms 2000) to experimentally test seedling drought tolerance in *S. obovata*, a highly self-fertilising species (Weller et al. 1996, 1998). We defined drought tolerance as the ability to survive periods of drought without a significant reduction in growth. Following the norm of reaction approach, siblings from three selfed maternal families, each from a different population, and two interpopulation crosses were grown under control and simulated drought conditions, and various metrics of seedling performance were quantified (e.g. life expectancy, total biomass, height, total leaf area, and leaf number). Comparing the performance of selfed and outcrossed progeny is a common concern of conservation managers (Edmands 2007; Maschinski et al. 2013), because small populations may be more likely to express inbreeding depression than large populations (Fenster and Dudash 1994; Dudash and Fenster 2000).

In addition to fitness metrics, we measured several morphological and physiological traits for insights into the mechanisms underlying drought tolerance or vulnerability, including rates of photosynthesis, conductance, transpiration, chlorophyll content, specific leaf area, and root:shoot biomass ratio, among others.

Our specific questions and predictions were as follows:

- (1) Can *S. obovata* seedlings tolerate simulated drought, and is there intraspecific variation in drought tolerance? Based on its restricted geographic range within mesic habitats (Price et al. 2012), we predicted that *S. obovata* would be susceptible to drought at the seedling stage, and that it would express low intraspecific variation in drought tolerance due to the low number of plants remaining.
- (2) Do physiological and morphological traits of *S. obovata* seedlings demonstrate phenotypic plasticity in response to simulated drought,

and do these patterns vary among progeny from different maternal families? Consistent with patterns reported for other *Schiedea* species (Norman et al. 1995; Culley et al. 2006), we predicted phenotypic plasticity in response to drought, genetic variation in the traits, and genetic variation in phenotypic plasticity.

- (3) Which functional traits promote tolerance to drought for *S. obovata* seedlings? We expected to detect traits associated with water uptake and water use efficiency (e.g. root:shoot biomass ratio and photosynthetic rate/transpiration) to promote tolerance to drought in *S. obovata* seedlings.
- (4) Do selfed versus crossed plants differ in their overall performance?

We predicted that *S. obovata* individuals produced via out-crossing would have greater overall performance (i.e. life expectancy and growth) than progeny from self-pollinated plants.

Materials and methods

Study system

As part of the Polynesia-Micronesia Biodiversity Hotspot (Myers et al. 2000), the Hawaiian flora is unique with high rates of endemism: 91% of angiosperm and 77% of fern and lycophyte taxa (Wagner et al. 1999; Palmer 2003). *Schiedea obovata* (Sherff) (Caryophyllaceae) is a short-lived endemic Hawaiian shrub that grows up to 1 m in height. Historically, *S. obovata* was scattered along ridges and slopes in mesic forests dominated by *Acacia koa* (Fabaceae) and *Metrosideros polymorpha* (Myrtaceae) throughout the Waianae Mountain Range on the island of Oahu, from 550 to 800 m above sea level (a.s.l.) and from areas with ca. 1300 to 1900 mm of rainfall per year (Wagner et al. 1999; TNC 2006). Over the past several decades, *S. obovata* has declined in population density and geographical distribution and is now restricted to the northern end of the Waianae Mountain Range (Keir and Weisenberger 2011), and to higher elevation sites that receive high annual rainfall (Loope and Giambelluca 1998). In 1991, *S. obovata* was federally listed as endangered (USFWS 1991), and the primary drivers implicated in its decline were habitat degradation by feral pigs, herbivory by slugs, invasive plants, trampling and over-collection by humans, consequences of small population size, and climate change (USFWS 1991, 2012). Currently, one population (NW Makaleha), at

the high end of the elevation gradient, of ca. 100 individuals (30 mature) and a few isolated individuals in surrounding areas (W Makaleha and Keawapilau) represent the extant, *in situ* distribution for this species. Though populations at the lower end of the elevation gradient are extirpated, one to several wild plants were collected from and subsequently represented through *ex situ* germplasm and *in situ* recovery efforts.

Greenhouse experiment

To generate the seeds used in this greenhouse experiment, fruits were collected from wild and outplanted plants in 2008; this represents the parental generation. The wild plants originated from three sites: West Makaleha (four parental plants represented in this study), Kahanahaiki (one parental plant represented), and Pahole Rim (one parental plant represented). The sites occur along an elevation gradient from Makaleha at the highest elevation (792 m a.s.l.), to Pahole at intermediate elevation (700 m a.s.l.), and Kahanahaiki at the lowest elevation (573 m a.s.l.), and they vary in climate, particularly in mean annual rainfall (wettest to driest: Makaleha, Pahole, Kahanahaiki, Figure S1). F_1 progeny were grown in the greenhouse to reduce maternal effects and were hand-pollinated (Weisenberger 2012) to produce either selfed F_2 progeny (from three source populations along the elevation gradient) or F_2 progeny from crosses between populations. One interpopulation cross was made between Makaleha and Pahole F_1 plants, and another between Makaleha and Kahanahaiki F_1 plants, to produce two crossed maternal families. Seeds were then dried to 20% RH and stored at -18°C between collection and germination.

In October 2016, a total of 326 seeds were sown on 1% water agar (Sigma-Aldrich A1296), and germination occurred over a period of 6–8 weeks. In total, 194 seeds germinated and reached the seedling stage. Following germination, seedlings were transplanted to 10 cm-diameter pots and moved to a greenhouse for the duration of the experiment. Once seedlings produced their first pair of true leaves, they were transplanted into pots of a volume of 3.79 L (1-gallon) filled with equal parts by volume of potting mix (Pro-Mix®, by Premier Co.) and black cinder, with a single application of slow-release fertiliser (Osmocote Plus 15-9-12, Scotts Co., Marysville, OH, USA) and placed under shade cages that admitted 50% of photosynthetically active radiation (PAR). Each 3.79 L

pot was filled with 1700 ± 50 g of this soil mixture prior to transplantation, to ensure equal amounts of soil across all replicates. Transplantation to the pots marks day zero (D0) of the experiment. On D14, seedlings were randomly assigned to two watering treatments that consisted of a control group that was watered to 100% field capacity (FC) daily, and a single pulse drought treatment that received no water for 14 days. Following the end of the pulse drought, all plants were watered daily before they were either harvested or subjected to a terminal drought. The pulse drought treatment was intended to mimic a potential future climate scenario in Hawaii, whereby plants are subjected to increasingly long droughts (Timm et al. 2015; Frazier 2016; Frazier and Giambelluca 2017) punctuated by more extreme precipitation events (Chu et al. 2010; Chen and Chu 2014; Zhang et al. 2016).

Five weeks after the end of the pulse drought, half of the plants in each treatment group (100% FC and pulse drought) were harvested. The remaining plants were subjected to a terminal drought and received no water until they died. The number of days until death while in a terminal drought was thus an estimate of life expectancy, the amount of time an organism is expected to remain alive. This design allowed us to assess whether plants subjected to well-watered or a pulse drought treatment varied in their life expectancy, an important fitness component. Plants were considered dead when the shoots were either leafless or the leaves brown, and the stem was brittle.

The full experimental design was two watering treatments (100% FC and pulse drought) $\times$ two fates (harvested or terminal drought) $\times$ five maternal families (3 selfed and 2 interpopulation crosses; see Table S1 for sample sizes for each maternal family).

Trait data collection

At the time of harvest, we measured leaf thickness (mm), seedling height (cm), the number of leaves, and the total leaf area (cm^2) for all individuals. Leaf thickness was measured with a digital micrometre (Fowler High Precision, Newton, MA, USA). All leaves per plant were photographed, and total leaf area was estimated using ImageJ v. 1.51q 18 (Abramoff et al. 2004). Stems, leaves, and roots were harvested and dried separately in an oven for 7–14 days at 60°C and weighed using an analytical balance to estimate total biomass; root:shoot ratio was calculated by dividing the root by shoot biomass. Specific leaf area (SLA) was calculated by

dividing the leaf area of the youngest fully mature leaf by its mass. Dried leaves were finely ground, and samples were sent to the University of Hawaii at Hilo Analytical Laboratory for leaf carbon and nitrogen analyses. We report the carbon to nitrogen ratio on a mass basis.

For a subset of plants in each treatment group, net photosynthesis, transpiration rate, and stomatal conductance were measured on the youngest fully expanded leaf per plant, within four days before the harvest. Instantaneous water use efficiency (WUE, $\mu\text{mol CO}_2 \text{ mmol}^{-1} \text{ H}_2\text{O}$) was calculated by dividing photosynthetic rate (A) by transpiration rate (E). Photosynthesis and transpiration rates were measured using a LiCOR 6400XT (Li-Cor, Lincoln, NE, USA) with PAR $1500 \mu\text{mol m}^{-2} \text{ s}^{-1}$, and CO_2 concentration set to $400 \mu\text{mol m}^{-2} \text{ s}^{-1}$. The observed average ($\pm\text{SE}$) leaf temperature was $28^\circ\text{C} \pm 2$, and average ($\pm\text{SE}$) relative humidity was $58\% \pm 9$. Stomatal conductance (hereafter, conductance) was measured using a steady state diffusion porometer (SC-1, Decagon, Pullman, Washington, USA). Chlorophyll content was measured with a SPAD 502 Plus Chlorophyll Metre (Spectrum Technologies, Inc, Aurora, IL, USA) for three haphazardly selected leaves and averaged.

For seedling that were not harvested and entered the terminal drought, we recorded the date of mortality for each individual. These data were then used to estimate and test for differences in life expectancy (i.e., days until death). To determine the critical threshold of water availability (below which plants die), volumetric soil water content (soil water content) of pots was measured on the day each plant died using a HydroSense II soil-moisture probe (Campbell Scientific, Logan, UT, USA). Soil water content was converted to soil matric potential using volumetric moisture as a conversion factor (Figures S2–S5). We also weighed the pots to calculate gravimetric water content compared to fully saturated pots at 100% field capacity (Sack 2004).

Data analysis

As the traits were not strongly correlated (Table S2), we performed univariate analyses instead of a MANOVA. To evaluate whether the performance of *S. obovata* seedlings varied among maternal families, due to watering treatment, and their interaction, we conducted two-way ANOVAs with total biomass, height, number of leaves, and leaf area, as the dependent variables in separate analyses. Maternal family was treated as a fixed factor in an ANOVA framework rather than a random factor in

a linear mixed model because the maternal families were selected to represent the historic distribution of *S. obovata* over an elevation gradient. These analyses determined whether *S. obovata* seedlings were drought tolerant (main effects of drought), whether there was genetic variation in seedling performance (maternal family main effects), and whether there was intraspecific variation in drought tolerance (drought $\times$ maternal family interactions). To test for potential differences in seedling life expectancy (i.e., days until death in terminal drought) by watering treatment and among maternal families, we used a parametric survival regression model (i.e., accelerated failure-time model). We used Akaike Information Criterion (AIC) to identify the distribution that best fit our data. Significant effects of watering treatment and maternal family on seedling life expectancy were assessed using analysis of deviance.

To determine whether the morphological (SLA, leaf thickness, root biomass, root:shoot ratio) and physiological (net photosynthesis, stomatal conductance, instantaneous WUE, transpiration, chlorophyll content, carbon to nitrogen ratio) traits were influenced by the watering treatment, maternal family, and their interaction, we conducted a two-way ANOVA for each trait separately. This statistical analysis allowed us to assess whether *S. obovata* seedlings exhibited plasticity in their traits in response to a drought (drought main effect), and whether there was intraspecific variation in phenotypic plasticity (drought $\times$ maternal family interaction).

To investigate how plant traits related to *S. obovata* seedling performance under drought, we used analysis of covariance (ANCOVA) models similar to those described above for drought tolerance analyses, but with the addition of each trait individually as a covariate. Positive relationships between traits and performance metrics (main effects or in an interaction) indicate that the traits are related to performance (i.e. that they have a functional role in determining growth and life expectancy) and for cases in which the relationship is positive in drought, that they are potential mechanisms of drought tolerance. When trait $\times$ drought interactions are significant, a positive relationship between the trait and performance metric under drought but not control conditions, indicates that the traits are mechanisms of drought tolerance but do not enhance performance under control conditions. Negative relationships between traits and performance metrics reveal that the traits

constrain performance. We conducted a similar analysis to test for interactions between traits and maternal family on seedling growth, however we did not test for a three-way interaction between maternal family, watering treatment, and the physiological traits due to low sample sizes.

To test whether outcrossed families had higher performance than selfed families [via heterosis (Charlesworth and Charlesworth 1987)], we conducted a two-way ANOVA on total biomass, height, number of leaves, and leaf area, separately, with breeding programme (selfed or crossed), watering treatment, and their interaction as the independent variables.

All statistical analyses were conducted in R v. 3.3.2 (R Core Team 2013). For the functional traits, we tested for significant pair-wise comparisons among populations using Tukey-Kramer HSD at $\alpha = 0.05$, which accounts for unequal sample sizes among groups. Total biomass and leaf area were square root-transformed to achieve normality prior to all ANOVAs and ANCOVAs. Interaction plots were generated using the 'lava' package. Remaining plots were created using the 'ggplot2' package and baseline plotting functions.

Results

Water availability

Soil water content (%) and soil matric potential (an index of available water) were relatively consistent over the duration of the experiment, i.e. there was no significant temporal effect (soil water content: $F_{1,90} = 0.90$, $P = 0.34$; soil matric potential: $F_{1,90} = 2.67$, $P = 0.11$, Figure S6). There were no significant differences in soil water content or soil matric potential at the time of death (terminal soil water content and terminal soil matric potential, respectively) between the treatments (control versus pulse drought) (soil water content: $F_{1,90} = 0.75$, $P = 0.39$; soil matric potential: $F_{1,90} = 1.20$, $P = 0.28$) or maternal families (soil water content: $F_{4,87} = 1.15$, $P = 0.34$; soil matric potential: $F_{4,87} = 2.04$, $P = 0.10$) (Figure S6), and this was consistent over the course of experiment ($P > 0.05$).

Seedling performance and drought tolerance

Of the 100 individuals subjected to the 14-day pulse drought, four died over the subsequent five weeks of the experiment. Half of the surviving 96 plants were

then harvested while the remaining half entered the terminal drought treatment 99% of seedlings that entered the terminal drought treatment died. Seedling growth was measured using four indicators: seedling height, number of leaves, total leaf area, and total biomass. The pulse drought had no significant effect on height, total leaf area, or the number of leaves (Figure 1; Table 1) but did significantly reduce total biomass by 38% (Figure 1; Table 1), indicating weak tolerance of drought. For plants in the terminal drought experiment, there was no main effect of prior watering treatment on life expectancy ($\chi^2_1 = 5.98$, $P = 0.84$) or terminal soil water content (volumetric water content: $F_{1,86} = 0.80$, $P = 0.37$; soil matric potential: $F_{1,90} = 0.75$, $P = 0.39$), indicating that the pulse drought treatment did not prime plants to have greater life expectancy in the terminal drought nor did it make them more susceptible to the terminal drought.

We detected no variation in life expectancy ($\chi^2_4 = 5.98$, $P = 0.2$) or terminal soil water content

($F_{4,86} = 1.15$, $P = 0.34$) among maternal families, indicating that there was no detectable intraspecific variation in these metrics of drought tolerance. While the maternal families differed significantly in total biomass, height, number of leaves, and total leaf area, revealing intraspecific variation in seedling growth (Table 1; Figure S7), there was no evidence of a watering treatment by family interaction on seedling height, biomass, number of leaves, and leaf area. These results indicate no genetic variation in growth due to the pulse drought (all $P > 0.05$, Table 1). We did, however, detect that the pulse drought influenced how the maternal families performed under the terminal drought (significant family $\times$ drought interaction on life expectancy: $\chi^2_4 = 12.25$, $P = 0.016$, Figure 2). Of the five maternal families, seedlings from the selfed Makaleha maternal family had significantly lower life expectancy under terminal drought following exposure to pulse drought; seedlings exposed to the pulse drought lasted nearly two weeks less in terminal

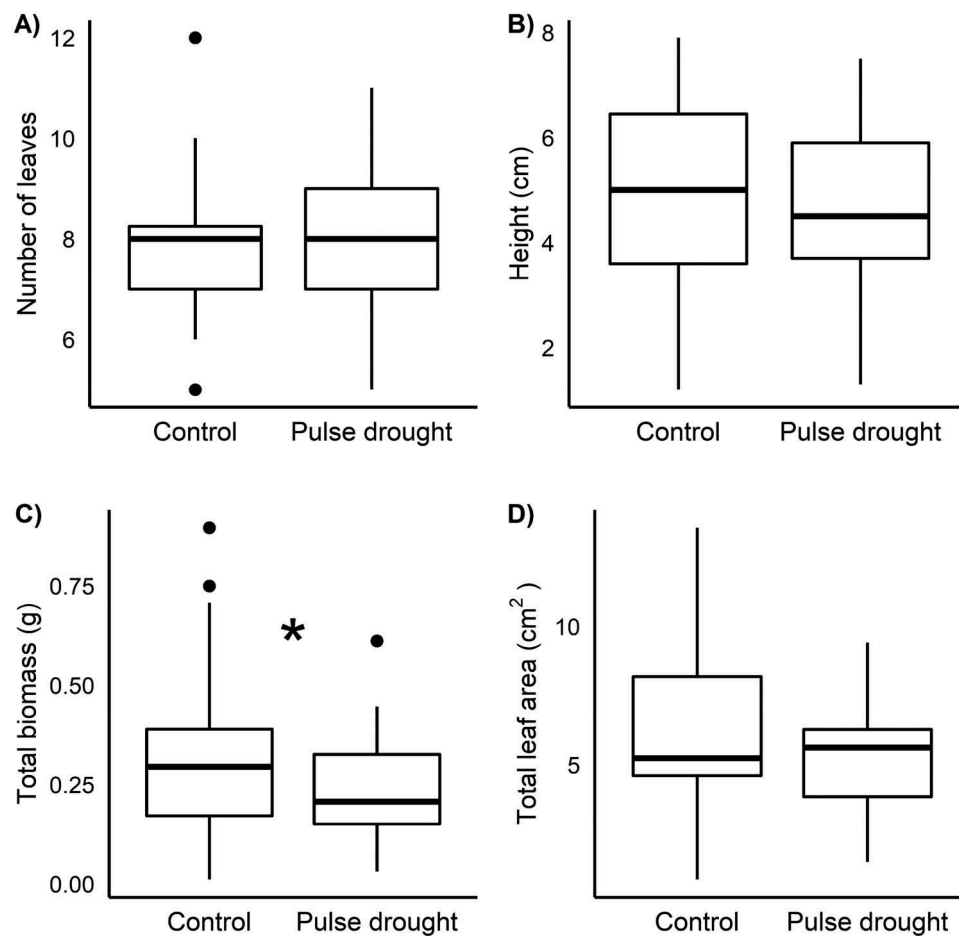


Figure 1. Effect of watering treatments (control versus pulse drought) on *Schiedea obovata* seedling size (total biomass, height, number of leaves, and total leaf area). Plants in the control group were watered daily to 100FC, while those in pulse drought received no water for 14 days. Means $\pm$ S.E. are shown for all traits. *indicates statistically significant differences among watering treatments at $\alpha = 0.05$.

Table 1. Influence of maternal family, watering treatment, breeding programme (selfed or crossed) (main effects) and interactions between treatment and maternal family or breeding programme, on *Schiedea obovata* seedling size. All test statistics and p-values were generated using ANOVA. Boldface indicates statistically significant effects ($\alpha \leq 0.05$).

Response variable	Explanatory variable(s)	F	df	P
Total biomass	Treatment	4.60	1,86	0.03
	Maternal family	11.42	4,86	<0.0001
	Breeding	10.73	1,92	0.001
	Treatment×Maternal family	1.17	4,86	0.33
	Treatment×Breeding	0.05	1,92	0.83
Height	Treatment	1.48	1,83	0.23
	Maternal family	15.49	4,83	<0.0001
	Breeding	2.76	1,89	0.10
	Treatment×Maternal family	0.11	4,83	0.98
	Treatment×Breeding	0.44	1,89	0.51
Total leaf area	Treatment	2.00	1,59	0.16
	Maternal family	4.85	4,59	0.002
	Breeding	7.10	1,65	0.01
	Treatment×Maternal family	0.17	4,59	0.95
	Treatment×Breeding	0.03	1,65	0.86
Number of leaves	Treatment	0.05	1,78	0.82
	Maternal family	2.59	4,78	0.04
	Breeding	1.89	1,84	0.17
	Treatment×Maternal family	0.48	4,78	0.75
	Treatment×Breeding	1.88	1,84	0.17

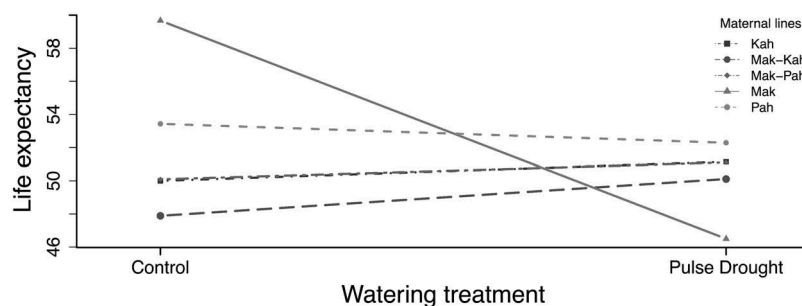


Figure 2. Interaction plot of maternal family and watering treatment on life expectancy of *Schiedea obovata* seedlings, which was significant at $\alpha = 0.05$. Abbreviations denote different maternal families: Mak = Makaleha, Mak/Kah = Makaleha and Kahanahaiki cross, Mak/Pah = Makaleha and Pahole cross, Kah = Kahanahaiki, Pah = Pahole.

drought than seedlings from the control watering treatment. The remaining maternal families had similar life expectancies following control or drought watering treatments, indicating limited intraspecific variation in tolerance to repeated drought events, in terms of seedling life expectancy.

Breeding programme (selfed vs. interpopulation progeny) significantly influenced seedling growth; progeny from crosses were 7.5 and 11% larger than selfed progeny in total biomass and leaf area, respectively, revealing evidence for possible hybrid vigour in *S. obovata* seedlings (Table 1). The influence of the watering treatments on the plant traits did not vary by breeding programme (Table 1). Breeding programme did not influence life expectancy ($\chi^2_1 = 2.45$, $P = 0.12$), nor did it vary in response to the watering treatment ($\chi^2_1 = 3.47$, $P > 0.05$).

Variation in functional traits

We detected little evidence for phenotypic plasticity in physiological or morphological traits in response to the pulse drought (Table 2). Only leaf C:N ratio (reduced by 4% compared to control plants) and SLA (increased by 9% compared to control plants) were significantly influenced by the pulse drought (Table 2; Figure 3(a, b)). The root:shoot ratios were marginally higher in pulse drought compared to control conditions (Table 2; Figure 3(c)).

There was significant variation in both physiological and morphological traits among maternal families (Table 2; Figures S8 and 9). While some of this variation was of small magnitude (e.g. SLA varied by only 5% among maternal families), other traits demonstrated considerable variation (e.g. 77% variation for total root biomass).

Table 2. Influence of watering treatment, maternal family, and the interaction between maternal family and watering treatment on traits in *Schiedea obovata* seedlings. All test statistics and p-values were generated using ANOVA. Boldface indicates statistically significant effects ($\alpha \leq 0.05$).

Response variable	Explanatory variable(s)	F	df	P
Photosynthesis	Treatment	0.20	1,59	0.65
	Maternal family	1.78	4,59	0.14
	Treatment×Maternal family	0.16	4,59	0.96
Transpiration	Treatment	0.07	1,59	0.80
	Maternal family	7.58	4,59	0.0001
	Treatment×Maternal family	0.11	4,59	0.98
Water use efficiency	Treatment	0.15	1,59	0.70
	Maternal family	3.91	4,59	0.01
	Treatment×Maternal family	0.22	4,59	0.93
Conductance	Treatment	1.77	1,60	0.19
	Maternal family	11.33	4,60	<0.0001
	Treatment×Maternal family	1.28	4,60	0.29
Chlorophyll content	Treatment	0.55	1,60	0.46
	Maternal family	15.36	4,60	<0.0001
	Treatment×Maternal family	0.13	4,60	0.97
Carbon:Nitrogen	Treatment	4.04	1,59	0.05
	Maternal family	6.30	4,59	0.0003
	Treatment×Maternal family	0.53	4,59	0.71
SLA	Treatment	4.18	1,59	0.05
	Maternal family	3.70	4,59	0.009
	Treatment×Maternal family	0.42	4,59	0.80
Leaf thickness	Treatment	0.24	1,60	0.63
	Maternal family	22.10	4,60	<0.0001
	Treatment×Maternal family	0.40	4,60	0.81
Root biomass	Treatment	0.33	1,86	0.57
	Maternal family	8.17	4,86	<0.0001
	Treatment×Maternal family	1.13	4,86	0.35
Root:shoot	Treatment	3.72	1,86	0.06
	Maternal family	1.26	4,86	0.29
	Treatment×Maternal family	1.34	4,86	0.26

Trait relationships to fitness

We observed significant main effects of traits on seedling growth as well as some interactions between physiological traits and the watering treatment, revealing linkages between traits and seedling drought tolerance (Table S3). Traits found to enhance seedling performance similarly under control and pulse drought conditions (i.e. no significant interaction) include photosynthesis, water use efficiency, chlorophyll content, and leaf C:N (Table S3; Figure 4). In contrast, conductance appeared to constrain performance as evidenced by significant negative effects on seedling height and leaf number (Table S3; Figure 4(h,i)).

In addition, we detected four significant interactions between physiological traits and watering treatment on seedling growth (Table S3; Figure 5). Chlorophyll content, photosynthesis, and conductance had positive or neutral effects on seedling size under both control and drought conditions, whereas leaf carbon to nitrogen ratio was the only trait that enhanced growth (in this case, number of leaves) under control conditions but constrained growth under drought.

There was also significant intraspecific variation in traits linked with seedling growth (Table S4, Figure S10). Chlorophyll content was the only trait that was significantly associated with all four growth performance metrics in distinct ways across the maternal

families (a significant trait × maternal family interaction). The associations were predominantly positive with the exception of the number of leaves, which increased with chlorophyll content for two of the families while decreasing or remaining constant for the remaining families. In addition to chlorophyll, the maternal families varied in several other trait-growth relationships, including: photosynthesis-biomass, WUE-biomass, conductance-height, C:N-leaf area, conductance-leaf number. Total biomass increased with photosynthesis and WUE for all families except the Makaleha/Pahole cross. With increasing stomatal conductance, the number of leaves decreased for all families except Pahole and had a neutral or negative effect on height for the remaining families. Higher leaf C:N ratios were associated with greater leaf area in 4 of the 5 maternal families; leaf area in Makaleha plants was insensitive to leaf C:N. The remaining traits-growth relationships were consistent across the maternal families.

Discussion

For future plant persistence, effective conservation strategies are needed to evaluate and ameliorate ongoing threats associated with global climate change. Rare, threatened, and endangered plant species are especially at risk because their

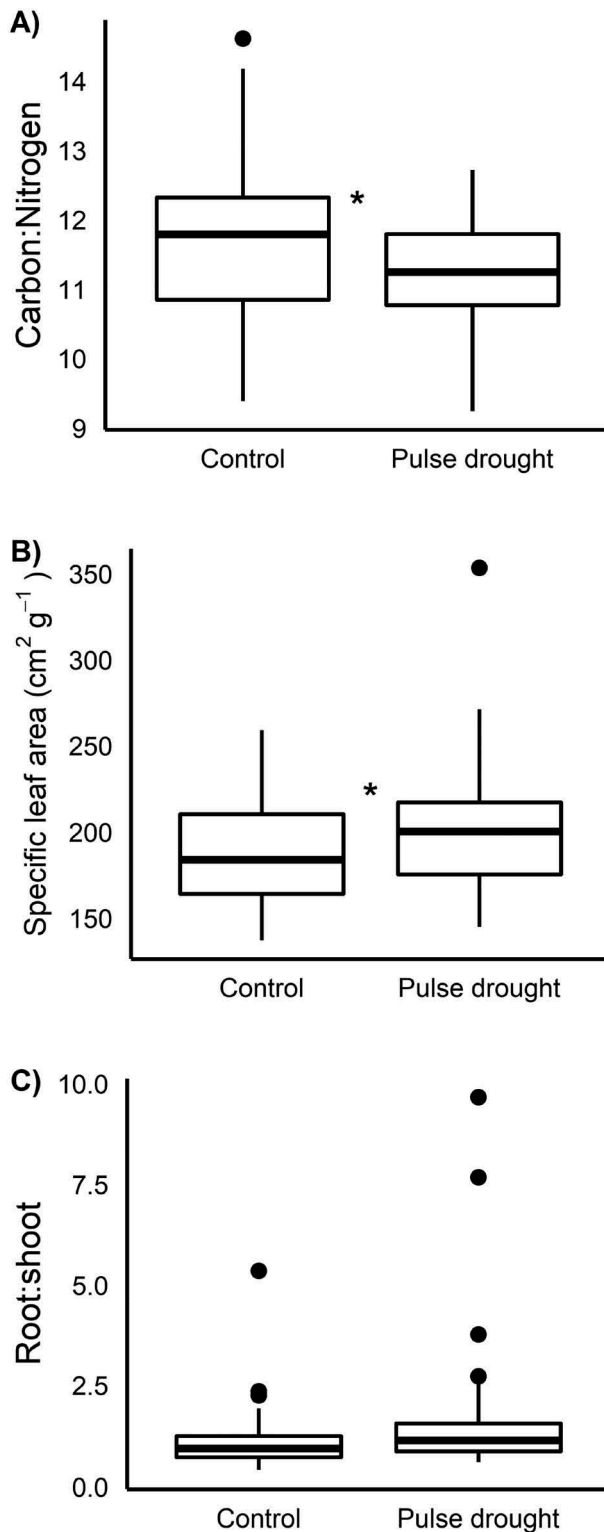


Figure 3. Effect of watering treatments (control versus pulse drought) on leaf nutrients (A) and morphology (B, C) of *Schiedea obovata* seedlings. Plants in the control group were watered daily to 100FC, while those in pulse drought received no water for 14 days. Means $\pm$ S.E. are shown for all traits. *indicates statistically significant differences among watering treatments at $\alpha = 0.05$. Root:shoot differences were marginally significant at $\alpha = 0.10$.

population sizes are small and may lack the genetic diversity necessary for adaptation (Leimu et al. 2006). To conserve rare and threatened species

requires intensive management strategies that may include controlled breeding (both within and among populations), mitigation of threats, and translocations. These strategies would benefit from identification of source populations and interpopulation crosses that are most resistant to abiotic stressors (Edwards 2015). In this study, we provide evidence of low to moderate drought stress, low phenotypic variation under drought stress, moderate genetic variation in traits, and implications of breeding programme for drought stress, in seedlings of *S. obovata*, part of a Hawaiian genus that is exceptionally vulnerable to climate change (Fortini et al. 2013). These results can inform management of this endangered species, as well as provide a framework for assessing intraspecific variation in stress tolerance of rare plants.

Evidence of drought stress

We defined drought tolerance as the ability to survive during periods of low moisture without a significant reduction in growth, and measured various metrics of seedling fitness under drought stress, including total leaf area, total biomass, number of leaves, height, and life expectancy. For most of these fitness metrics, we detected similar performance in drought-stressed and control plants, suggesting relatively high drought tolerance. However, total biomass, perhaps the most comprehensive fitness metric, was significantly reduced (by nearly 40%) in drought-stressed compared to control plants, indicating weak tolerance within this fitness metric. These findings partially supported our prediction that *S. obovata* would be sensitive to drought, resulting from its narrow distribution range in terms of elevation and precipitation, and suggests that this species may be vulnerable to future reductions in mean annual precipitation that are predicted across the Hawaiian archipelago (Timm et al. 2015; Frazier and Giambelluca 2017).

A reduction in total biomass under drought indicates that *S. obovata* seedlings shift their carbon allocation patterns to maintain their height, leaf area, and number of leaves despite the considerable reduction in total biomass (a reduction of nearly 40%). Because seedling size is an important predictor of short- and long-term survival as well as future size of juvenile life stages (Gilbert et al. 2001), reductions in seedling biomass under drought warrant further investigation. In this study, the reduction in total biomass did not result from the production of thinner leaves, less root biomass, or total leaf area (see section on Trait

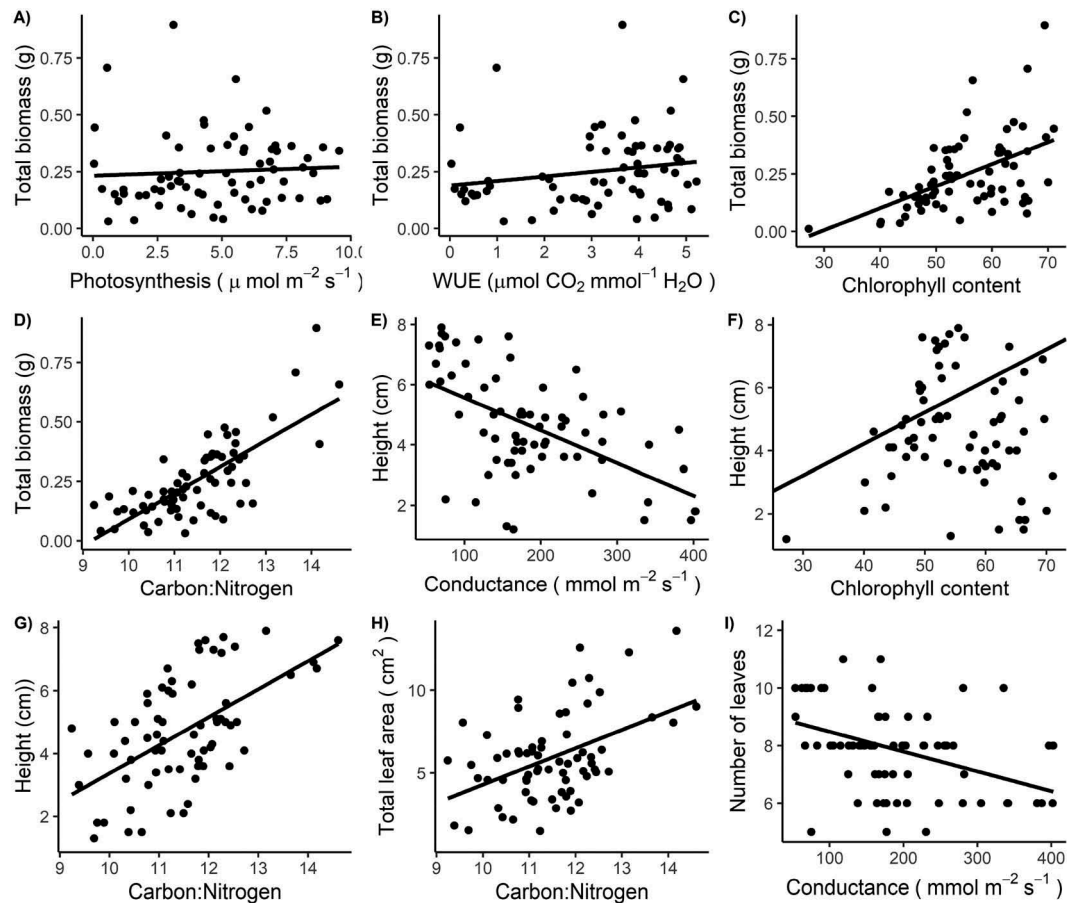


Figure 4. Effects of leaf physiology on seedling size for *Schiedea obovata*. All significant main effects at $\alpha = 0.05$ are pictured. Lines represent regression relationships on untransformed variables.

plasticity), and is likely due to a reduction in stem density, which we did not estimate (leaf and stem biomass were not measured separately). Stem (or wood) density is closely associated with water transport in woody species (Poorter and Markesteijn 2008), and a reduction in stem density could prove deleterious for *S. obovata* as forests in Hawaii' become drier (Chu et al. 2010; Chen and Chu 2014; Zhang et al. 2016). Thus, *S. obovata* seedlings under drought exhibited a shift in their carbon allocation strategy, reducing investment in stems while maintaining the ability to intercept light, i.e. height, leaf area, and leaf number (Ackerly and Bazzaz 1995; Valladares and Niinemets 2007; Duursma et al. 2012).

Intraspecific variation in fitness metrics and drought tolerance

In addition to investigating overall drought stress of this endangered species, we sought to quantify the extent of intraspecific variation in performance across maternal families. While we detected significant variation in seedling growth across

maternal families, we also observed considerable consistency among the maternal families in their responses to drought, suggesting low intraspecific variation in seedling drought tolerance. Local adaptation to microsite climatic conditions may underlie observed differences in the maternal families, but whether the results detected in this study indicate local adaptation requires additional genetic sampling across a greater climatic gradient and further experimentation, ideally in the field. Given that the focal species is critically endangered, the sample sizes used in the present study are necessarily limited. Despite this limitation, our findings provide useful insight regarding the extent of intraspecific variation in drought tolerance for an endangered Hawaiian endemic.

Trait plasticity

With the exception of leaf C:N ratio, which was slightly lower under drought, we found no evidence that *S. obovata* seedlings exhibit plasticity in their physiology in response to drought stress, at least not in the maternal families examined. There was

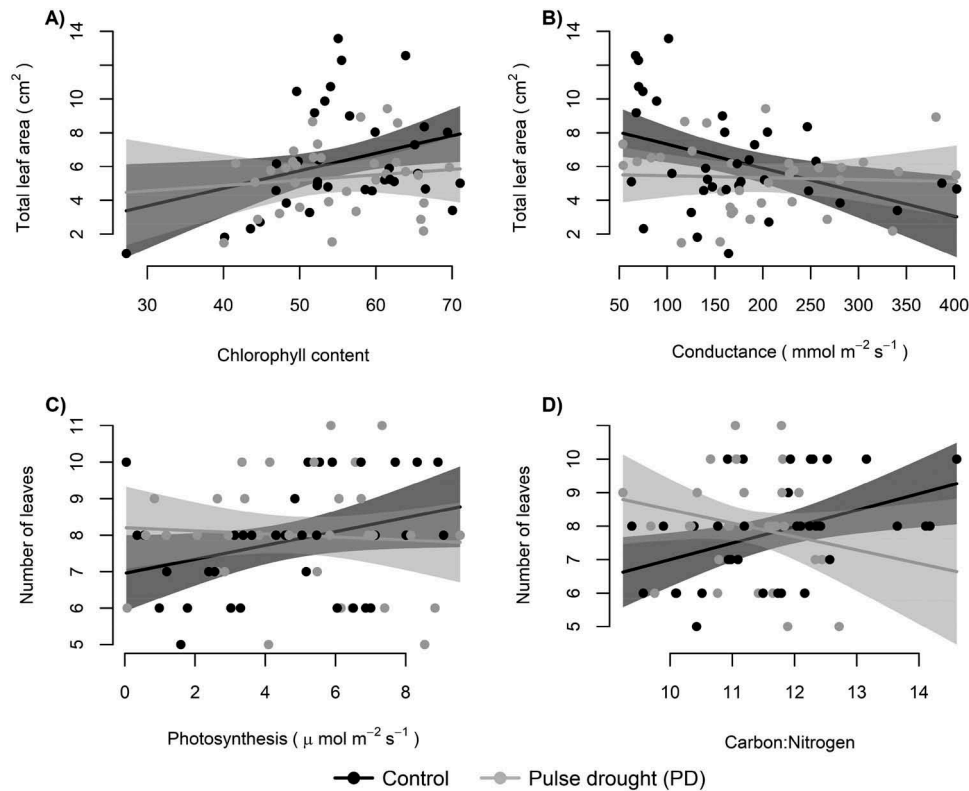


Figure 5. Interactive effects of leaf physiology on seedling size for *Schiedea obovata* under a pulse drought (PD, grey circles) and a control treatment watered daily (control, black circles). All significant two-way interactions are shown, see Supplementary material for nonsignificant interactions. Coloured bands represent 95% confidence intervals for pulse drought (grey) and control (black), respectively.

some evidence of plasticity in morphological traits, as exposure to drought triggered seedlings to produce leaves with a greater specific leaf area but equal thickness compared to those that were watered daily. These findings were contrary to our prediction that seedlings would express significant plasticity, based on a previous study conducted on congeneric species (Norman et al. 1995), in which plasticity in photosynthesis and conductance in response to soil fertility was tested within selfed and crossed maternal families from a single population of a largely outcrossing species. To our knowledge, our study is the first to examine physiological and morphological trait plasticity in response to simulated drought in a *Schiedea* species, and the lack of plasticity observed could indicate that *S. obovata* is unable to modulate gas exchange dynamics and leaf morphology as a mechanism of drought tolerance. In general, phenotypic plasticity is thought to buffer against changing environmental conditions (Nicotra et al. 2010) and may enable the long-term persistence of remnant individuals or populations, especially at the edges of their geographic range (Matesanz et al. 2010). However, a lack of plasticity may not necessarily be maladaptive for rare and endangered species (Alpert

and Simms 2002; Ghalambor et al. 2007) and may be selected against when the energetic costs of being plastic outweigh the benefits, or when the environment is highly stable (see review by Reed et al. 2011).

Traits conferring drought tolerance

We tested for trait $\times$ drought interactions, predicting to find a subset of traits associated with enhanced performance under drought conditions, but with a weaker relationship with performance under control conditions (in other words, a more positive slope for plants in drought compared to control). This result would allow us to identify traits that operate specifically as drought tolerance mechanisms. Previously, Dudley (1996) has used experimental manipulations to demonstrate that the influence of leaf gas exchange rates on leaf size varies with moisture availability, and that this variation in size influences reproductive fitness. Similar findings have been reported with regards to genetic differentiation among populations in photosynthetic physiology in response to other abiotic gradients, including temperature, light, and elevation, which influences temperature, precipitation, and potential evapotranspiration [see review by Arntz

and Delph (2001) and references therein], although relatively few of these studies have been conducted on seedlings. We detected four significant interactions, but for only one of these, was the slope more positive for the drought than control conditions, and that was stomatal conductance. Under control conditions, seedling size declined as stomatal conductance increased, while no trend was observed for stomatal conductance under drought. These findings suggest that gas exchange via stomata may be carefully regulated under periods of drought, and that this regulation allows *Schiedea* seedlings to maintain their size during unfavourable conditions. Our results are most consistent with an isohydric water use strategy, where plants tolerate drought by decreasing leaf size and reducing stomatal conductance when water is limiting (Cohen 1970; Dudley 1996; McDowell et al. 2008). Although seedlings under drought did not have lower conductance on average than those under well-watered conditions, we did detect a fitness advantage (greater leaf area) conferred by maintaining low conductance under drought.

For the other three significant interactions, the positive relationship between the traits and seedling performance were more positive under control than the pulse drought conditions, revealing them to be liabilities under drought. For example, an increase in the C:N ratio was linked with an increase in total leaf number for control plants, but under pulse drought, it was the opposite; a higher C:N ratio lead to lower leaf number. Previous studies have found that plants that allocate more biomass to roots than shoots have a greater capacity to capture and use nitrogen from the soil, resulting in higher shoot nitrogen concentrations and $\delta^{15}\text{N}$ (Lloret et al. 1999). *Schiedea obovata* seedlings exposed to a pulse drought appear to follow this pattern, shifting allocation towards roots and increased leaf nitrogen content, at the expense of biomass. Our findings indicate that trait functionalities are contingent upon moisture availability such that high carbon to nitrogen ratios enhance seedling performance under well-watered conditions while a greater relative investment in nitrogen enhances seedling performance under drought.

When we examined trait $\times$ maternal family interactions, chlorophyll content was consistently identified as a trait whose influence on seedling size varied among the maternal families. Chlorophyll content is often associated with an increase in photosynthetic rates (Björkman 1981), as observed in the present study, but it may also influence fluorescence parameters (i.e. the ability to tolerate high light stress)

under drought stress (Li et al. 2006). The observed intraspecific variation in chlorophyll–growth relationships may result from the carbon costs associated with chlorophyll synthesis (De Vries 1975), or from underlying variation in resource allocation strategies, such as changes in leaf thickness, which influences chlorophyll content (Neilsen et al. 1995). To a lesser extent, stomatal conductance, photosynthesis, water use efficiency, and leaf C:N ratio were also associated with one or more metrics of seedling size in distinct ways among the maternal families. In general, however, the directions of the trait effects on size were consistent among the families although they varied in magnitude. Intraspecific variation in traits may explain differences in seedling performance among the families. For example, chlorophyll content was higher in plants from Makaleha compared to those from Kahanahaiki, which represent the extremes of size in the present study, from smallest to largest, respectively. Understanding the extent to which this intraspecific variation trait–growth relationships among maternal families is linked with drought tolerance is outside the scope of the present study but would be informative for future conservation efforts in the context of increasing regional drought. To fully understand how variation in growth sensitivity influences performance under drought (a three-way interaction: trait $\times$ treatment $\times$ maternal family), would require much larger sample sizes than were available for this endangered endemic shrub.

Evidence of outcrossing benefits

We detected evidence of greater performance in seedlings resulting from crosses compared to selfed plants, consistent with our prediction. Seedlings produced from interpopulation crosses were slightly larger than progeny produced by self-pollination in terms of biomass and leaf area. However, we did not observe that crossing enhanced drought tolerance, for the few crossed groups examined here. Our results are also somewhat consistent with those reported by Weisenberger (2012), which detected heterosis in terms of seeds per fruit but not growth (based on height and number of branches) and life expectancy. Differences in results could be attributed to variation in the environment in which the plants were grown (field versus greenhouse), the methods used, or the focal fitness metrics. Maschinski et al. (2013) were the first to document a benefit of mixing propagules from multiple populations for population persistence following rare and extreme climatic events. Because small populations are often more

likely to experience inbreeding depression (Dudash and Fenster 2000), supplementing populations with individuals from other populations can increase population size and potentially increase genetic diversity (Edmands 2007). For outcrossing species, maintaining adequate population sizes and genetic diversity can be achieved via outcrossing. However, for selfing *Schiedea* taxa, including *S. obovata* (Weller et al. 1996, 1998), while planting individuals from crosses among populations rather than within populations may enhance offspring vigour on short time scales, occasional outcrossing is unlikely to regularly provide sufficient heterosis to maintain selection for outcrossing (Lande and Schamske 1985), i.e. it may not be self-sustaining. Therefore, the long-term benefits of supplementing populations with progeny from crosses between populations remain unknown.

Future breeding programmes for *S. obovata* could thus consider crossing genotypes from within the historic distribution range of this species to enhance seedling performance. However, these decisions should be weighed against the potential negative effects of diluting locally adapted genomes and disrupting historically favourable genotype $\times$ environment interactions (Maschinski et al. 2013). Given the observed intraspecific variation in seedling life expectancy under terminal drought for selfed progeny, more work is needed to determine whether interpopulation crosses should be pursued as conditions become drier overall, and more severely punctuated by extreme drought events, as is projected for Hawaii (Timm et al. 2015; Frazier and Giambelluca 2017). Furthermore, reciprocal transplant studies are needed to determine how progeny from the lower end of the species elevational gradient will perform at the higher end of the gradient and beyond, if managers wish to pursue assisted translocations as a tool for ameliorating the effects of drought on this species. Determining the best balance between any population variation in drought tolerance and potential benefits of establishing populations with interpopulation crossed progeny, could be evaluated based on the suitable habitat identified for species recovery efforts.

Conclusions

We found evidence of low intraspecific variation in the ability to tolerate drought, and moderate drought tolerance overall, for seedlings of an endangered, endemic Hawaiian shrub. Additionally, there was some variation in life expectancy following repeated drought events. We also identified various morphological and

physiological traits that significantly influenced seedling performance and warrant further investigation. Our findings suggest that *S. obovata* seedlings may be susceptible to reductions in rainfall within Hawaii, where climate change scenarios project more severe drought (Timm et al. 2015; Frazier and Giambelluca 2017). We also detected significant genetic variation in seedling growth, morphology, and ecophysiology, highlighting that even rare, endangered species can still harbour considerable genetic variation in traits influencing fitness. Assessments of phenotypic plasticity and stress tolerance in vulnerable life stages may thus provide managers and restoration scientists with better predictions regarding how threatened and endangered species will cope with climate change.

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Disclosure statement

No potential conflict of interest was reported by the authors.

Notes on contributors

Andrea Westerband is interested in plant adaptations to limiting resources, with a particular emphasis on variation in physiological traits across resource gradients.

Lalasia Bialic-Murphy studies plant-environmental interactions and the effects of resource limitation and mutualism disruption on native plant population dynamics.

Lauren A. Weisenberger focuses on the recovery of rare plant species and the restoration of their habitats in Hawaii and other Pacific islands.

Kasey E. Barton is interested in seedling stress tolerance, particularly for island endemic plants facing drought, herbivory, and competition.

ORCID

Andrea C. Westerband  <http://orcid.org/0000-0003-4065-9689>

Data availability

Trait data will be made available online at DRYAD. We will not release any spatial data (GPS coordinates or other identifiers) because accessing the populations that provided the genetic material for this study is strictly prohibited by the US Fish and Wildlife Service.

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